SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.1: Introduction to Chemistry — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Introduction to Chemistry
Driving Question	DRIVING QUESTION: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Defining Chemistry: What Happened to the Chapati Dough?**	Students observed the anchoring phenomenon: raw chapati dough (flour + water + fat) is soft, pale, and pliable, but after cooking on a hot pan it becomes firm, golden-brown, and crisp — and cannot be reversed to raw dough. Teacher should display or demonstrate both states side-by-side. Students also examined other everyday matter changes: a rusting panga, dissolving sugar in chai, burning firewood. Key observation to elicit: the cooked chapati is permanently different from the raw dough — something chemically changed. Prompt students: 'Can you turn the cooked chapati back into raw dough?' (Expected: No.) 'What do you think caused the change?'	Chemistry is the scientific study of matter — its properties, composition, structure, and the changes it undergoes. Matter is anything that has mass and occupies space (e.g., chapati dough, water, air, soil). Changes to matter are classified as: (1) Physical changes — reversible, no new substance formed (e.g., dissolving sugar in water, melting fat); (2) Chemical changes — usually irreversible, new substance(s) formed with different properties (e.g., chapati dough cooking, rusting iron, burning wood). Evidence of chemical change includes colour change, gas production, heat/light release, and irreversibility. The cooked chapati is the anchoring phenomenon for the entire sub-strand — return to it in every lesson.	The chapati dough undergoes a chemical change during cooking: heat energy breaks and reforms chemical bonds in the starch and protein molecules of the flour, producing new substances (dextrins, melanoidins via Maillard reaction) with entirely different properties. This is why the change is irreversible. Teacher note: Students commonly confuse physical and chemical changes — use the reversibility test as a quick heuristic ('Can you get back exactly what you started with?'). Cold-call at least 3 students to classify a given change (e.g., chopping onions = physical; frying an egg = chemical). Connect to DQ: Chemistry is the tool that explains why the chapati changed — and why that knowledge matters.
Lesson 2 **Branches of Chemistry: Mapping the Science to Our World**	Students examined a set of Kenyan real-world scenarios (printed cards or written on the board) and attempted to sort them before instruction: (a) a Kericho tea farmer testing soil pH; (b) a Nairobi hospital pharmacist preparing a painkiller; (c) a KPLC engineer working on a nuclear	Chemistry is divided into specialised branches: (1) Organic Chemistry — study of carbon-containing compounds (e.g., carbohydrates in ugali, fats in cooking oil, medicines); (2) Inorganic Chemistry — study of non-carbon compounds (e.g., fertiliser salts, water-treatment chemicals,	No branch of Chemistry operates in isolation — the chapati phenomenon, for example, involves organic chemistry (starch and protein reactions), physical chemistry (heat energy transfer), and biochemistry (digestion). Teacher note: Use the sorting activity debrief to correct misconceptions —

	energy report; (d) a cook at a Mombasa hotel preserving fish; (e) a Kisumu water-treatment officer purifying Lake Victoria water. Students recorded which scenarios felt 'similar' and why. Teacher should allow 3–4 minutes of small-group sorting with WAIT TIME before cold-calling 3 groups to share their reasoning.	metals); (3) Physical Chemistry — study of energy and physical properties in chemical processes (e.g., heat released when burning charcoal); (4) Analytical Chemistry — identification and measurement of substances (e.g., KEBS testing unga for contamination); (5) Biochemistry — chemistry of living organisms (e.g., how the body digests githeri); (6) Industrial/Applied Chemistry — large-scale chemical processes (e.g., EABL brewing, Bamburi cement production); (7) Nuclear Chemistry — changes in atomic nuclei (e.g., nuclear energy, medical imaging). Each branch explains a different aspect of why matter behaves as it does.	students often think Chemistry only happens in laboratories. Emphasise that every Kenyan farmer, cook, nurse, and engineer uses chemical knowledge daily, even if they do not call it 'Chemistry.' Pedagogical tip: Create a class 'Branches of Chemistry' anchor chart on the wall and add examples from each subsequent lesson to reinforce cumulative understanding.
Lesson 3 **Chemistry in Agriculture and Food — How Chemistry Feeds Kenya**	Students observed or read about two contrasting Kenyan farm scenarios: (A) a Rift Valley maize farm using DAP fertiliser and pesticide spray, producing high yields; (B) a farm using no inputs, struggling with poor soil and pest damage. Where possible, teacher should display a bag of DAP fertiliser, a pesticide container label, and a packet of iodised salt for students to read and handle. Students recorded: What chemicals are mentioned on the labels? What do they claim to do? What warnings are given? Allow 5 minutes of label reading before cold-calling 4 students to share one observation each.	Chemistry is applied in Kenyan agriculture and food in three main ways: (1) Fertilisers (e.g., DAP, CAN, urea) — change soil chemistry to supply nitrogen, phosphorus, and potassium that plants need to synthesise proteins and carbohydrates. Overuse causes soil acidification and water pollution — a chemistry consequence. (2) Pesticides and herbicides — chemically target harmful organisms (e.g., organophosphates disrupt insect nerve chemistry) while ideally sparing crops and humans; misuse leads to chemical residues in food. (3) Food chemistry — preservation (salting, pickling, use of sodium benzoate), fortification (iodised salt prevents goitre; fortified unga supplies iron and vitamins), and food safety testing (KEBS analytical chemistry). Understanding chemistry helps farmers and consumers make safer, more productive choices.	The same chemistry that makes a pesticide effective (disrupting an organism's biochemical pathways) can make it dangerous to humans if misused — dosage and application method are chemistry decisions. Teacher note: Connect directly to the DQ ('safer, smarter choices') — a farmer who understands fertiliser chemistry avoids over-application that acidifies soil and pollutes Kisumu's water. Connect back to the chapati: the wheat flour in chapati is produced by a farm that used fertiliser chemistry. Pedagogical tip: If a student's family farms, invite them (with sensitivity) to share what inputs their family uses — ground the lesson in lived experience.
Lesson 4 **Chemistry in Medicine, Pharmaceuticals, and Manufacturing**	Teacher displays or describes 3 common Kenyan medicine scenarios: (a) a packet of paracetamol (Panadol) with dosage instructions; (b) a newspaper headline about counterfeit medicines seized in Nairobi; (c) a chlorine tablet used to purify drinking water. Students examine the paracetamol label and answer: What is the recommended dose? What happens if you	(1) Drugs interact with the body at a molecular level — they work by binding to specific receptor molecules or blocking certain chemical reactions in cells. The correct dose delivers a therapeutic effect; an overdose overwhelms the body's chemistry and causes toxicity. (2) Pharmaceutical chemistry involves synthesising, purifying, and testing drug	The concept of dosage is a chemistry concept — the body is a chemical system, and every substance introduced (food, medicine, poison) produces a dose-dependent chemical response. Teacher note: This lesson directly sets up Lessons 7–9 on drug/substance use. Establish early: 'The difference between a medicine and a poison is often the dose' (Paracelsus

	take more? Why does the label say 'keep out of reach of children'? Allow 4 minutes of label analysis, then cold-call 3 students. Teacher should also briefly reference Kenyan pharmaceutical companies (e.g., Dawa Limited, Universal Corporation) as local manufacturers applying chemistry.	compounds for safety and efficacy — this is why counterfeit medicines (lacking the correct chemical compounds) are dangerous. (3) Manufacturing chemistry: household and industrial products (soap, disinfectant, paint, cement, plastics) are produced through controlled chemical reactions. Examples: Bidco processing cooking oil from palm/sunflower seeds (hydrogenation); Kenpoly manufacturing plastic containers (polymerisation). (4) Water purification uses chlorine chemistry — chlorine reacts with pathogens to render water safe.	principle — state it clearly). Cold-call 2 students: 'Give me one example of a chemical manufactured in Kenya and say which branch of Chemistry it involves.' Correct misuse of the word 'chemical' as meaning only dangerous substances — everything is made of chemicals.
Lesson 5 **Chemistry in Other Industries: Nuclear, Sports, Entertainment & Energy**	Students read or listen to 4 short Kenyan-context vignettes: (A) Eliud Kipchoge's marathon performance and sports nutrition (carbohydrate loading, electrolyte drinks); (B) Kenya's exploration of nuclear energy as part of Vision 2030; (C) a Nairobi DJ using dry ice (solid CO_2) for stage effects; (D) Kenya's geothermal energy production at Olkaria, Naivasha. Students predict: 'Which of these involves Chemistry? How?' Allow 3 minutes, then cold-call 4 students (one per vignette). Expect students to struggle with nuclear and entertainment — this productive struggle is intentional.	Chemistry operates across diverse industries: (1) Nuclear Chemistry — energy is released when atomic nuclei split (fission) or join (fusion); Kenya is exploring nuclear energy as a clean power source (Vision 2030). Nuclear chemistry also underlies medical imaging and cancer treatment (radiotherapy). (2) Sports and Nutrition Chemistry — athletes like Kipchoge rely on carbohydrate (glycogen) chemistry for energy; electrolytes (sodium, potassium ions) maintain muscle chemistry; performance-enhancing drugs are banned because they illegally alter body chemistry. (3) Entertainment Chemistry — dry ice sublimation, fireworks (rapid oxidation reactions), stage lighting (photochemistry). (4) Energy Chemistry — geothermal (Olkaria), biogas from organic waste, and solar energy all involve chemical or physical energy conversion. Chemistry underpins every energy source Kenya uses.	Chemistry is not confined to laboratories or factories — it is present in every human activity. Teacher note: The nuclear energy point often surprises students; clarify that nuclear chemistry involves changes inside atomic nuclei (not just atomic electron changes as in ordinary chemistry) — this is a conceptual boundary worth making explicit. For sports nutrition, connect to biochemistry (Lesson 2). For Olkaria geothermal, connect to physical chemistry (heat and energy). Pedagogical tip: Ask students to name one industry in their home county and challenge them to find the Chemistry in it — this builds schema for the DQB update.
Lesson 6 **Careers in Chemistry and Gender Stereotyping**	Teacher presents profiles of 4 real or realistic Kenyan chemistry-related professionals: (1) a female KEBS food technologist inspecting unga in Eldoret; (2) a male nurse administering IV drugs at KNH; (3) a female environmental chemist monitoring pollution in Nairobi River; (4) a male farmer in Meru using foliar fertilisers. Before discussion, students respond to: 'Which of these jobs do you consider a	Chemistry underpins a wide range of Kenyan careers: food technologist, pharmacist, chemical engineer, environmental scientist, nurse/doctor, agricultural officer, laboratory technician, forensic scientist, cosmetologist, brewer/distiller, geothermal engineer, and teacher. All genders are equally capable of pursuing any chemistry career — historical and cultural stereotypes (e.g., 'girls are not	Gender stereotyping in science careers is itself a social phenomenon that Chemistry education can challenge — when students see themselves as future chemists, they invest more in learning. Teacher note: Handle this topic sensitively but directly. If a student expresses a gender stereotype, use a Socratic approach: 'What evidence supports that belief? Can you think of a Kenyan example that challenges it?' Do not

	'chemistry career'? Which could a woman do? Which could a man do?' Collect anonymous responses (show of hands or written) to surface stereotypes. Allow 2 minutes, then cold-call 5 students to share reasoning.	good at science'; 'lab work is for men') are not supported by evidence and limit Kenya's development. Kenyan women scientists (e.g., Prof. Miriam Were — public health; Dr. Lydia Ndirangu — chemistry education) have made significant contributions. The Kenya Constitution (Article 27) and CBC values of respect and social justice support gender equity in STEM.	single out students. Pedagogical tip: Connect to the DQ — 'making smarter choices' includes choosing a career based on interest and ability, not gender. This lesson also reinforces that Chemistry is a living, human discipline practised by real Kenyans.
Lesson 7 **Meaning of Drug, Prescription, Dosage and Substance Use**	Teacher displays or describes 5 common Kenyan substances and asks students to classify each as 'drug,' 'food,' 'both,' or 'neither,' and to justify: (a) paracetamol tablet; (b) cup of strong chai (tea with caffeine); (c) chang'aa (illicit alcohol); (d) cigarette; (e) uji (porridge). Allow 3 minutes of individual written responses, then cold-call 5 students. Expect disagreement on chai and uji — this productive disagreement motivates the need for precise definitions. Teacher should also display a sample prescription form and an OTC medicine packet side-by-side.	Key definitions students must know verbatim: (1) Drug — any substance that, when taken into the body, alters its normal functioning (physiological or psychological). (2) Prescription — a written instruction from a licensed medical professional authorising a patient to receive a specific drug at a specific dose. (3) Dosage — the measured quantity of a drug to be taken at one time or over a period, calculated to produce a therapeutic effect without toxicity. (4) Over-the-counter (OTC) medicine — drugs safe enough for self-administration without a prescription (e.g., paracetamol, antacids). (5) Prescription-only medicine — drugs requiring a doctor's prescription due to risk of misuse or serious side effects (e.g., antibiotics, opioids). (6) Substance use — taking any drug or psychoactive substance. (7) Substance abuse — using a substance in a way that is harmful to oneself or others, including misuse of prescription drugs or use of illegal substances.	The chemistry principle underlying all drug concepts is dose-response: every substance produces a biological effect proportional to the amount taken, and at some dose, every substance becomes harmful (even water in extreme excess). Teacher note: Caffeine in chai is technically a drug — this deliberately challenges students' assumptions and reinforces that 'drug' is a scientific term, not a moral judgment. The moral and legal dimensions are separate from the chemical reality. Pedagogical tip: Revisit Lesson 4's paracetamol example — same molecule, discussed now with richer vocabulary (dosage, prescription, OTC). Connect to upcoming Lessons 8–9 on specific substances.
Lesson 8 **Effects of Drug and Substance Use: Tobacco — Chemistry of an Irreversible Harm**	Teacher presents a labelled diagram or list of the chemical compounds found in tobacco smoke (nicotine, carbon monoxide, tar, formaldehyde, benzene — all named), and displays or describes two lung images: one healthy, one with chronic smoker's damage. Students observe and record: (1) Which of these chemicals have you heard of before, and in what context? (2) What do you predict each chemical does to the body? Allow 4 minutes of individual written predictions, then cold-call 4 students. Teacher notes: formaldehyde is a preservative; benzene is a solvent —	'Natural' does not mean 'safe' — tobacco is a natural plant, yet its chemical compounds cause specific, largely irreversible physiological damage: (1) Nicotine — addictive stimulant; binds to nicotinic acetylcholine receptors in the brain, causing dopamine release and physical dependence. Withdrawal is a chemical process. (2) Carbon monoxide (CO) — binds to haemoglobin with 200× greater affinity than oxygen, reducing blood oxygen-carrying capacity; causes chronic hypoxia. (3) Tar — complex mixture of carcinogenic compounds that coat lung	Tobacco illustrates the sub-strand's driving question most directly: knowing the chemistry of tobacco enables smarter, safer choices. Teacher note: Connect to the anchoring phenomenon — just as the chapati undergoes an irreversible chemical change when cooked, lung tissue undergoes irreversible chemical changes from tar exposure. This parallel is pedagogically powerful and should be stated explicitly. Avoid stigmatising students who may have family members who smoke; frame the lesson around chemical knowledge, not moral judgment. Cold-call 3

	students recognising these from other contexts demonstrates Chemistry's cross-domain nature.	alveoli, reducing gas exchange surface area, causing chronic bronchitis and lung cancer. (4) Formaldehyde and benzene — both carcinogens causing DNA damage. The damage is largely irreversible because lung tissue (alveoli) and DNA mutations cannot be fully repaired. Peer pressure is a social factor that initiates substance use; chemistry explains why stopping is physically difficult (addiction chemistry).	students: 'Name one chemical in tobacco smoke, the branch of Chemistry that studies it, and one body system it damages.' This integrates Lessons 2, 4, and 7.
Lesson 9 **Consumer Protection and the Safe Learning Environment**	Students examine (physically or via description) 3 Kenyan consumer products: (a) a packet of Omo detergent with a KEBS mark; (b) a bottle of mineral water with a KEBS certification sticker; (c) a counterfeit medicine packet (without KEBS/PPB mark — teacher should prepare a mock example). Students are asked: 'What information on these labels is chemistry-based? What protects you as a consumer? What is missing from the counterfeit product that makes it dangerous?' Allow 4 minutes of label analysis and written responses, then cold-call 4 students.	(1) Consumer protection in Kenya is enforced by KEBS (Kenya Bureau of Standards) for goods and PPB (Pharmacy and Poisons Board) for medicines — both use analytical chemistry to verify product composition and safety. (2) Preservative chemistry: food preservatives (e.g., sodium benzoate, ascorbic acid/Vitamin C) slow spoilage by inhibiting microbial growth or oxidation — they are safe at regulated doses but harmful above regulatory limits (dose-response, revisited). (3) Laboratory safety rules are themselves consumer/user protection: correct labelling, storage of chemicals, use of PPE, no eating in labs, safe disposal of chemicals — all are chemistry-informed safety practices that translate to real-world chemical handling. (4) A safe learning environment in Chemistry requires: mutual respect, freedom to ask questions without ridicule, equal participation regardless of gender, and physical safety (following lab rules).	Consumer protection is applied analytical and regulatory chemistry — chemists design the standards that protect Kenyan families. Teacher note: The connection between lab safety rules and real-world chemical safety is explicit: the same principle (know what a chemical is, handle it correctly, use the right amount) applies whether the student is in a school lab or a Nairobi market buying medicine. Pedagogical tip: Use this lesson to establish or reinforce classroom/lab safety norms for the rest of the Chemistry course. Connect to the DQ: KEBS and PPB exist because Chemistry knowledge protects people — learners who understand Chemistry can read labels, recognise fakes, and make safer choices.
Lesson 10 **Project: Poster on Drug & Substance Use — Unit Synthesis & Final Explanation**	Students present their completed posters on drug and substance use (prepared in groups) to the class. Each poster must include: a definition of drug/substance abuse; one specific substance (e.g., tobacco, alcohol, miraa/khat) with its chemical compounds and effects; connection to at least one branch of Chemistry; a consumer protection or safety message; and a Kenya-specific statistic or example. Teacher circulates during presentations, prompting with questions: 'Which branch of Chemistry explains this	Synthesis of the full sub-strand: (1) Chemistry is the scientific study of matter and its changes — physical (reversible, no new substance) and chemical (usually irreversible, new substance formed). (2) The branches of Chemistry (organic, inorganic, physical, analytical, biochemistry, industrial, nuclear) each explain a different dimension of the matter around us. (3) Chemistry is applied in agriculture (fertilisers, pesticides), medicine (drugs, dosage, pharmacology), manufacturing (FMCG products, cement, plastic), energy	The poster project is a performance task that assesses all sub-strand learning outcomes simultaneously: knowledge of chemistry concepts, ability to apply them to real Kenyan contexts, communication skills, and the values of responsibility and social justice (gender equity in careers, consumer rights). Teacher note: During presentations, use the Driving Question Board to show students how their early questions (Lesson 1) are now fully answered — this metacognitive closure is essential in the CBE model. Assess using the KICD-aligned

	effect?' / 'What does the chemistry tell us about why this is hard to stop?' / 'How does this connect to our driving question?' Audience students complete a peer-feedback form.	(geothermal, nuclear, biogas), sports, and entertainment. (4) Every manufactured product and medicine involves chemistry — consumer protection (KEBS, PPB) uses analytical chemistry to keep Kenyans safe. (5) Drugs alter body chemistry; dosage determines therapeutic vs. harmful effects; addiction is a chemical process; substance abuse causes largely irreversible chemical damage. (6) Chemistry careers are open to all genders; stereotypes must be actively challenged. The chapati dough that began the unit (an irreversible chemical change) now represents the full explanatory power of Chemistry: matter changes, and understanding those changes saves lives.	rubric: accuracy of chemistry content, Kenya-specific application, visual clarity, and group participation. After presentations, return to the chapati phenomenon one final time: 'We started by asking why chapati dough changed permanently. Now you know: it's Chemistry — and Chemistry explains, protects, and empowers us every day.'